

CHAPTER 4

Chemical Bonding and Molecular Structure

Covalent Bond, Electrovalent Bond, Lattice Enthalpy and Octet Rule

- Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is: NH_3 , AlCl_3 , BeCl_2 , CCl_4 , PCl_5 (2023)
a. 1 b. 3 c. 2 d. 4
- Which of the following species contains equal number of σ and π -bonds? (2015 Re)
a. $(\text{CN})_2$ b. $(\text{CH})_2(\text{CN})_2$
c. HCO_3^- d. XeO_4
- Which one of the following molecules contains no π bond? (2013)
a. SO_2 b. NO_2 c. CO_2 d. H_2O
- Which of the following is electron-deficient? (2013)
a. $(\text{CH}_3)_2$ b. $(\text{SiH}_3)_2$ c. $(\text{BH}_3)_2$ d. PH_3

Bond Parameters and Dipole Moment

- Which of the following molecules has "NON ZERO" dipole moment value? (2024 Re)
a. CCl_4 b. HI c. CO_2 d. BF_3
- Arrange the following compounds in increasing order of their solubilities in chloroform:
 NaCl , CH_3OH , cyclohexane, CH_3CN (2024 Re)
a. $\text{NaCl} < \text{CH}_3\text{CN} < \text{CH}_3\text{OH} < \text{Cyclohexane}$
b. $\text{CH}_3\text{OH} < \text{CH}_3\text{CN} < \text{NaCl} < \text{Cyclohexane}$
c. $\text{NaCl} < \text{CH}_3\text{OH} < \text{CH}_3\text{CN} < \text{Cyclohexane}$
d. $\text{Cyclohexane} < \text{CH}_3\text{CN} < \text{CH}_3\text{OH} < \text{NaCl}$
- Match List-I with List-II: (2024 Re)

	List-I Molecule		List-II Bond enthalpy (kJmol^{-1})
A.	HCl	I.	435.8
B.	N_2	II.	498
C.	H_2	III.	946.0
D.	O_2	IV.	431.0

Choose the correct answer from the options given below:

- a. A-III, B-IV, C-I, D-II b. A-IV, B-I, C-III, D-II
c. A-IV, B-III, C-II, D-I d. A-IV, B-III, C-I, D-II

- Identify the correct answer. (2024)
a. Dipole moment of NF_3 is greater than that of NH_3 .
b. Three canonical forms can be drawn for CO_3^{2-} ion
c. Three resonance structures can be drawn for ozone.
d. BF_3 has non-zero dipole moment.
- The correct order of dipole moments for molecules NH_3 , H_2S , CH_4 and HF , is: (2023-Manipur)
a. $\text{CH}_4 > \text{H}_2\text{S} > \text{NH}_3 > \text{HF}$ b. $\text{H}_2\text{S} > \text{NH}_3 > \text{HF} > \text{CH}_4$
c. $\text{NH}_3 > \text{HF} > \text{CH}_4 > \text{H}_2\text{S}$ d. $\text{HF} > \text{NH}_3 > \text{H}_2\text{S} > \text{CH}_4$
- The correct order of bond angles in the following compounds/species is: (2022 Re)
a. $\text{CO}_2 < \text{NH}_3 < \text{H}_2\text{O} < \text{NH}_4^+$
b. $\text{H}_2\text{O} < \text{NH}_3 < \text{NH}_4^+ < \text{CO}_2$
c. $\text{H}_2\text{O} < \text{NH}_4^+ < \text{NH}_3 < \text{CO}_2$
d. $\text{H}_2\text{O} < \text{NH}_4^+ = \text{NH}_3 < \text{CO}_2$
- Which of the following molecules is non-polar in nature? (2021)
a. CH_2O b. SbCl_5 c. NO_2 d. POCl_3
- Which of the following set of molecules will have zero dipole moment? (2020)
a. Boron trifluoride, hydrogen fluoride, carbon dioxide, 1, 3-dichlorobenzene
b. Nitrogen trifluoride, beryllium difluoride, water, 1, 3-dichlorobenzene
c. Boron trifluoride, beryllium difluoride, carbon dioxide, 1, 4-dichlorobenzene
d. Ammonia, beryllium difluoride, water, 1, 4-dichlorobenzene
- Which one of the following pair of species have the same bond order? (2017-Delhi)
a. N_2 , O_2^- b. CO , NO c. O_2 , NO^+ d. CN^- , CO
- Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statement is false? (2016 - I)
a. The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 is larger than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3
b. The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 , the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 , and the $\text{H}-\text{O}-\text{H}$ bond angle in H_2O are all greater than 90° .
c. Then $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is larger than the $\text{H}-\text{C}-\text{H}$ bond angle in CH_4
d. The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is smaller than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3

15. Which of the following molecules has the maximum dipole moment? (2014)

a. CH_4 b. NH_3 c. NF_3 d. CO_2

VSEPR Theory

16. Match List I with List II. (2024)

List I (Compound)		List II (Shape/geometry)	
A.	NH_3	I.	Trigonal Pyramidal
B.	BrF_5	II.	Square Planar
C.	XeF_4	III.	Octahedral
D.	SF_6	IV.	Square Pyramidal

Choose the correct answer from the options given below:

- a. A-III, B-IV, C-I, D-II
b. A-II, B-III, C-IV, D-I
c. A-I, B-IV, C-II, D-III
d. A-II, B-IV, C-III, D-I

17. Match List - I with List - II: (2022 Re)

List-I (molecules)		List-II (shape)	
A.	NH_3	I.	square pyramidal
B.	ClF_3	II.	trigonal bipyramidal
C.	PCl_5	III.	trigonal pyramidal
D.	BrF_5	IV.	T-shape

Choose the correct answer from the options given below:

- a. A-III, B-IV, C-I, D-II
b. A-II, B-III, C-IV, D-I
c. A-III, B-IV, C-II, D-I
d. A-IV, B-III, C-I, D-II

18. Amongst the following, which one will have maximum 'lone pair-lone pair' electron repulsions? (2022)

a. XeF_2 b. ClF_3 c. IF_5 d. SF_4

19. Match List-I with List-II.

List-I		List-II	
(A)	PCl_5	(i)	Square pyramidal
(B)	SF_6	(ii)	Trigonal planar
(C)	BrF_5	(iii)	Octahedral
(D)	BF_3	(iv)	Trigonal bipyramidal

Choose the correct answer from the options given below.

(2021)

- a. A-ii B-iii C-iv D-i
b. A-iii B-i C-iv D-ii
c. A-iv B-iii C-ii D-i
d. A-iv B-iii C-i D-ii

20. Identify the wrongly matched pair. (2020-Covid)

Molecule **Shape or geometry of molecule**

- a. SF_6 Octahedral
b. BeCl_2 Linear
c. NH_3 Trigonal pyramidal
d. PCl_5 Trigonal planar

21. Identify the incorrect statement related to PCl_5 from the following: (2019)

- a. Three equatorial P-Cl bonds make an angle of 120° with each other
b. Two axial P-Cl bonds make an angle of 180° with each other
c. Axial P-Cl bonds are longer than equatorial P-Cl bonds
d. PCl_5 molecule is non-reactive

22. The species, having bond angles of 120° is (2017-Delhi)

a. BCl_3 b. PH_3 c. ClF_3 d. NCl_3

23. Among the following which one is a wrong statement? (2016 - II)

- a. SeF_4 and CH_4 have same shape
b. I_3^+ has bent geometry
c. PH_3 and BiCl_3 do not exist
d. $p\pi - d\pi$ bonds are present in SO_2

24. Which of the following pairs of ions is isoelectronic and isostructural? (2016-II)

a. CO_3^{2-} , NO_3^- b. ClO_3^- , CO_3^{2-}
c. SO_3^{2-} , NO_3^- d. ClO_3^- , SO_3^{2-}

25. Predict the correct order among the following: (2016 - I)

- a. Lone pair — bond pair > bond pair — bond pair > lone pair — lone pair
b. Lone pair — lone pair > lone pair — bond pair > bond pair — bond pair
c. Lone pair — lone pair > bond pair — bond pair > lone pair — bond pair
d. Bond pair — bond pair > lone pair — bond pair > lone pair — lone pair

26. Maximum bond angle at nitrogen is present in which of the following? (2015)

a. NO_2^+ b. NO_3^- c. NO_2 d. NO_2^-

27. In which of the following pairs, both the species are not isostructural? (2015)

a. Diamond, Silicon carbide b. NH_3 , PH_3
c. XeF_4 , XeO_4 d. SiCl_4 , PCl_4^+

28. Which of the following pairs of ions are isoelectronic and isostructural? (2015)

a. SO_3^{2-} , NO_3^- b. ClO_3^- , SO_3^{2-}
c. CO_3^{2-} , SO_3^{2-} d. ClO_3^- , CO_3^{2-}

29. Which one of the following species has planar triangular shape? (2014)

a. NO_3^- b. NO_2^- c. CO_2 d. N_3^-

30. Which of the following is a polar molecule? (2013)

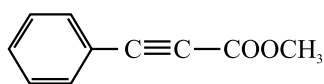
a. BF_3 b. SF_4 c. SiF_4 d. XeF_4

31. XeF_2 is isostructural with (2013)

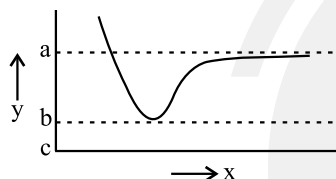
a. SbCl_3 b. BaCl_2 c. TeF_2 d. ICl_2^-

Valence Bond Theory, Hybridisation

32. Identify the incorrect statement about PCl_5 . (2024 Re)
 a. PCl_5 possesses two different Cl – P – Cl bond angles
 b. All five P – Cl bonds are identical in length
 c. PCl_5 exhibits sp^3d hybridisation
 d. PCl_5 consists of five P – Cl (sigma) bonds
33. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are: (2021)
 a. sp^3 and 6
 b. sp^2 and 6
 c. sp^2 and 8
 d. sp^3 and 4
34. How many (i) sp^2 hybridised carbon atoms and (ii) π bonds are present in the following compound? (2020-Covid)

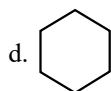
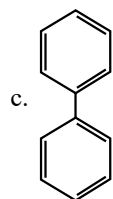
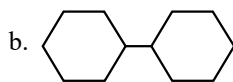
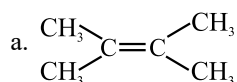


- a. 8, 6
 b. 7, 6
 c. 8, 5
 d. 7, 5
35. The potential energy (y) curve for H_2 formation as a function of internuclear distance (x) of the H atoms is shown below. (2020-Covid)



The bond energy of H_2 is

- a. $\frac{(c-a)}{2}$
 b. $\frac{(b-a)}{2}$
 c. $(c-a)$
 d. $(b-a)$
36. The hybridisation of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are: (2016-II)
 a. sp , sp^3 and sp^2
 b. sp^2 , sp^3 and sp
 c. sp , sp^2 and sp^3
 d. sp^2 , sp and sp^3
37. In which of the following molecules, all atoms are coplanar? (2016 - II)



38. The correct geometry and hybridization for XeF_4 are (2016 - II)
 a. Octahedral, sp^3d^2
 b. Trigonal bipyramidal, sp^3d
 c. Planar triangle, sp^3d^3
 d. Square planar, sp^3d^2
39. Which of the following organic compounds has same hybridisation as its combustion product (CO_2)? (2014)
 a. Ethyne
 b. Ethene
 c. Ethanol
 d. Ethane

MOT, Bonding in Some Homonuclear Diatomic Molecules

40. The correct order of energies of molecular orbitals of N_2 molecule is: (2023)
 a. $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$
 b. $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
 c. $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
 d. $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
41. Which one of the following statements is **incorrect** related to Molecular Orbital Theory? (2023-Manipur)
 a. The p^* antibonding molecular orbital has a node between the nuclei.
 b. In the formation of bonding molecular orbital, the two electron waves of the bonding atoms reinforce each other.
 c. Molecular orbitals obtained from $2p_x$ and $2p_y$ orbitals are symmetrical around the bond axis.
 d. A p -bonding molecular orbital has larger electron density above and below the internuclear axis.
42. Which amongst the following is incorrect statement (2022)
 a. O_2^+ ion is diamagnetic.
 b. The bond orders of O_2^+ , O_2 , O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively
 c. C_2 molecule has four electrons in its two degenerate π molecular orbitals.
 d. H_2^+ ion has one electron.
43. Identify a molecule which does not exist. (2020)
 a. Li_2
 b. C_2
 c. O_2
 d. He_2
44. Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory? (2019)
 a. O_2
 b. N_2
 c. C_2
 d. Be_2
45. Consider the following species: (2018)
 CN^+ , CN^- , NO and CN
 Which one of these will have the highest bond order?
 a. NO
 b. CN^-
 c. CN
 d. CN^+
46. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is: (2015 Re)
 a. $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$
 b. $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 c. $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
 d. $\text{O}_2 > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^-$

